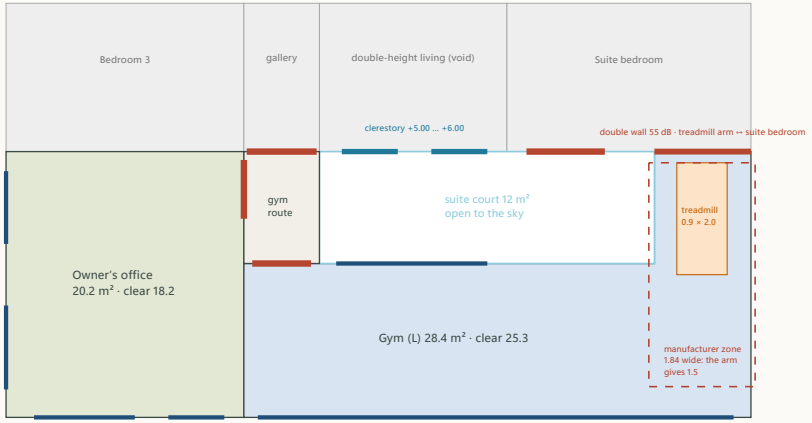
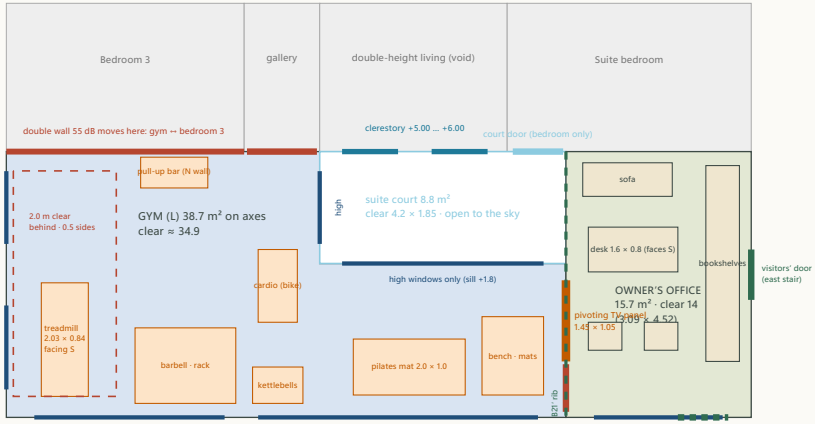
South wing study · gym west + south, office east, the court in the middle · and an external stair at the south-east · Taif / Palm House V7.1

Owner's decisions 1 Nov 2026, APPLIED the same day (docs/PENDING_CHANGES.md): the swap with the suite court 2.5 m clear (y 19.25–21.9), the east enclosed stair tower continuing to the roof, the internal roof stair dropped; the pivoting TV panel NOT applied (it does not fit beside the office door in the 2.1 m shared wall: see the report) · this study keeps its original comparison · every size is a study assumption; the structure is a re-trace of the load paths, not a calculation (X5 stays open)



CURRENT V7.1 · north up · 1 m = 0.74 cm



SWAPPED (owner's idea) · north up · 1 m = 0.74 cm

1 · The swapped wing: what it gives

- **Gym, an L:** the whole west part (x 0–5.6, 5.6 × 4.75 on axes) plus the south strip (x 5.6–10, 2.75 deep): **38.7 m² on axes, ≈ 34.9 m² clear** (today 28.4 / 25.3). Entered as today from the gallery end (x 4.3–5.55, y 19.25, sliding); windows west (two) and the south band x 0.5–9.5; onto the court **high-level windows only** (sill +1.8): the court stays private.
- **The treadmill fits with its manufacturer zone:** Life Fitness T5 class (assumed), 2.03 × 0.84 m, 2 m clear behind and 0.5 m each side (ASTM F2115 / the manual, assumed) → a zone 1.84 × 4.03 m; placed along the **west outer wall line** (the slab edge over the foyer's load-bearing west wall: the stiffest strip), facing south to the window band; it fits. Today's arm is 1.5 m clear: the owner's 0.3 m sides fit, the manufacturer's 1.84 m does not.
- **Also in the gym:** pull-up bar on the north wall, barbell + rack and kettlebells at the south-west, one cardio bike by the gallery door, the pilates mat and a bench in the south strip; doors open; the walkway from the gallery door to the office door runs along the court's walls (≥ 1.1 m).
- **Suite court:** x 5.6–10, 2.0 deep: **8.8 m² on axes, 4.2 × 1.85 clear** (today 12 m²); enclosed on four sides: the living's clerestory wall and the suite bedroom north, the gym west (x 5.6) and south (y 21.25), the office east (x 10); reached from the suite bedroom only (a 0.9 m sliding door at x 9.05–9.95, the only part of the bedroom's south wall that touches the court). Its floor drain west as today.
- **Owner's office, east:** x 10–13.311 × 4.75: **15.7 m² on axes, 3.09 × 4.52 = 14 m² clear** (today 20.2 / 18.2): a desk facing the south window, a sofa on the suite wall, shelves east. Doors: from the gym's south strip (0.85, lockable from the office side) and the visitors' door to the external stair (east wall, or south). The office window south only (x 10.5–12.8); an east window only if the stair is not on the east.

2 · The pivoting TV panel in the shared gym / office wall (x 10, y 21.55–23)

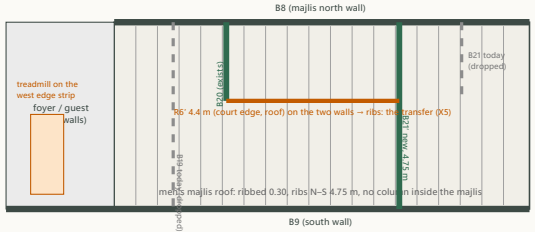
- **Opening:** 1.45 wide × 1.05 high (sill +0.95, head +2), 0.3 m south of R6's bearing on this wall, a lintel over it; the office door south of it (y 23.05–23.9).
- **Panel:** 0.18 m thick on a vertical centre pivot (the wall is 0.20: the panel's faces stand 0.01 proud each side, trimmed by a 15 mm stone architrave both sides). A welded steel frame, two 12.5 mm skins (fibre-cement or MDF), 100 mm mineral wool, a 55" TV (≈ 1.23 × 0.71) recessed flush in one face with a soundbar below it, the other face finished as the wall (or a second screen later). ≈ 75 kg incl. the TV; a floor pivot bearing + a top guide; cables through the hollow pivot post (slip ring or a service loop for 180°).
- **Sealed in both positions:** the panel turns 180° and fills the opening in either position; compression gaskets on all four edges are pressed by two cam locks (top / bottom) so there is no open gap — while turning, the half-width (0.72) swings into both rooms: keep 0.8 m free of furniture each side.
- **Expected sound reduction (assumed Rw):** the solid 200 mm plastered block wall ≈ 50 dB; the sealed panel ≈ 36 dB; the wall with the panel (1.52 m² in 9.63 m²) ≈ **43.2 dB**, i.e. 6.8 dB less than the solid wall; with worn gaskets (panel 30 dB) ≈ 37.8 dB. Enough between a gym and an office (normal speech inaudible, loud music audible); nothing like the 55 dB double wall a bedroom needs. The acoustic engineer confirms.

3 · Noise and privacy next to the master bedroom

- **Today** the treadmill arm shares 1.725 m of wall (y 19.25) with the suite bedroom (double wall, 55 dB) and the gym's high windows look onto the court outside the bedroom's door. **Swapped:** the **office** shares that wall (quiet: a single 200 mm wall suffices); the gym is 3.35 m away across the court; the gym now touches **bedroom 3** (y 19.25, x 0–4.25): the double wall moves there (same length, same cost). Gym noise still reaches the court and, through the clerestory windows, the double-height living (as today).

South wing study · checks passed (7)

4 · Structure re-traced (X5 stays open) · 5 · The clerestory light · 6 · Comparison with the current V7.1 south wing

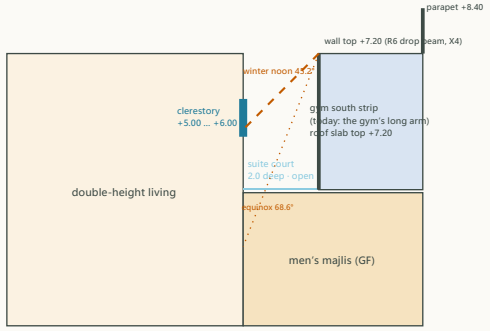


the men's majlis roof and the swapped walls · 1 m = 0.52 cm

4 · The load paths of the swapped wing

Wall	Stands on	Change
x 5.6, y 19.25–21.25 (gym west ↔ court)	thickened rib **B20** (exists today under the same wall)	as today
x 10.0, y 19.25–24.0 (court ↔ office, gym ↔ office)	a thickened rib **B21′** on x 10.0 for the full 4.75 m (today B21 sits on x 11.586 over 2.0 m): parallel to the ribs, direct and cheap	moved rib
y 21.25, x 5.6–10.0 (court ↔ gym south strip)	crosses the ribs, 4.4 m, nothing below: a **light partition** (100 mm, high windows) as recommended today for the 7.7 m gym north wall	partition
y 19.25, x 0–13.311 (bedroom 3 / living / suite bedroom south walls)	beam **B8** on the men's majlis north wall	as today
x 0, y 24, x 13.311	outer walls on the ground-floor outer walls (foyer west wall, majlis south and east walls)	as today
x 4.25 (today's office / gym route wall on B19)	no wall any more: B19 not needed, or kept as a stiffening rib under the treadmill if the treadmill moves to x 4.25	dropped

- **Roof over the wing:** the gym's west part (x 0–5.6, 4.75 span) ribbed 0.30 N–S like today's office roof; the south strip (2.75 span) solid 0.20 as today; the office (3.3 wide) solid 0.20 spanning **east–west** between the x 10 wall and the east outer wall. **R6′**, the court's south roof edge, shrinks from 6.0 to 4.4 m (≈ −27 % load) but is still a wall-on-rib transfer: **X5 is lighter, not closed.**
- **Treadmill:** on the slab's west edge strip over the foyer's load-bearing west wall (0.5 m off the wall face) — the stiffest place in the wing; alternative: keep a thickened rib under it (B19 moved to x ≈ 1.05). Floating floor and pads in any case (X6).
- **Light roof option:** an insulated steel deck on steel beams over the gym's west part and the office (the south strip stays concrete under the PV zone Z4): ≈ 2–3 kN/m² less on the majlis ribs, easier for X5; PV fixings and the parapet detail to design. The owner decides whether the roof may look lighter from the roof terrace.
- **The engineer must confirm:** the ribs B20 / B21′ carrying the walls and R6′ (sizes, deflection); the majlis roof's vibration under the treadmill (home gym live load, SBC 301); the lintel over the pivot opening; the roof option; the east outer wall taking the office roof.



N–S section at x ≈ 7.3 through the clerestory and the court, looking west · IDENTICAL in both layouts · 1 m = 0.5 cm

5 · The clerestory light

- **North–south (the section):** both layouts keep the court 2.0 m deep in front of the clerestory windows and a wall top at +7.20 behind it (the gym's south strip = today's long arm). The winter noon sun (45.2°) reaches **81 %** of the window height in both; the equinox sun 100 %; the sky seen from a seated eye in the living 100 %. **No change.**
- **East–west (the difference):** the court's east wall moves from x 11.586 (the treadmill arm) to x 10 (the office), 1.4 m east of the east clerestory window instead of 2.99 the morning sun from the south-east reaches that window a little later in winter:

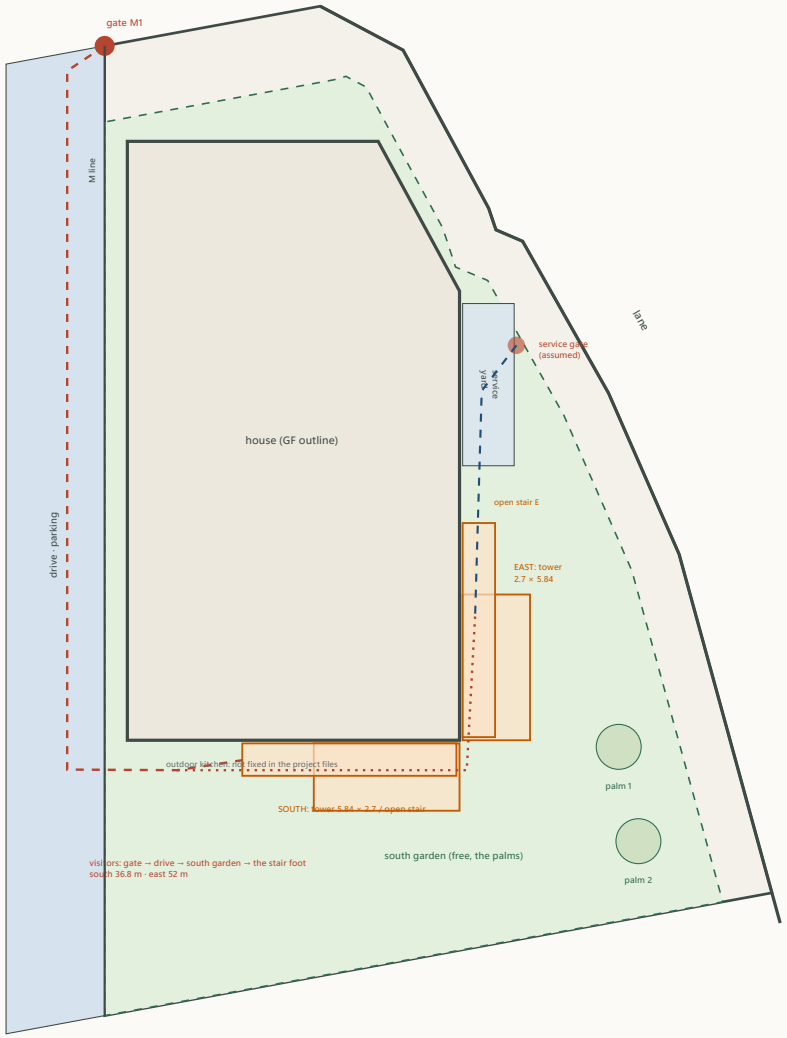
21 Dec, solar hour	altitude	east of the plan's south	today (wall x 11.586)	swapped (wall x 10)
9:00	27.3°	68.3°	shaded	shaded
10:00	36.5°	56.2°	sun	shaded
11:00	42.8°	40.3°	sun	shaded
12:00	45.2°	21.4°	sun	sun

- The west clerestory window (x 6.0–7.0) is 3 m from the office wall: as today's distance to the arm, no change. The office's own light does not depend on the court (south window); the gym's south part has the south band. **Conclusion: the office reduces the east window's early winter sun by about an hour; noon and afternoon light are unchanged.**

6 · Comparison with the current V7.1 south wing

Item	Current V7.1	Swapped (owner's idea)
Gym size, treadmill fit	28.4 m² on axes (≈ 25.3 clear); the treadmill arm 1.5 clear: 0.3 m sides only, the manufacturer's 1.84 m does not fit	38.7 m² (≈ 34.9 clear); the treadmill with 2.0 m behind and 0.5 m sides fits on the stiff west edge; everything else fits with doors open
Suite court size, privacy	12 m²; overlooked by the gym's north high windows and the arm's wall; the gym route door beside it	8.8 m² (4.2 × 1.85 clear): smaller; four walls, high windows only, one door from the bedroom: more private
Noise next to the master bedroom	the treadmill arm on the bedroom wall (double wall 55 dB)	the office on the bedroom wall (quiet); the gym 3.35 m away; the double wall moves to bedroom 3
Structure / X5	R6 6.0 m on two walls on ribs B20 / B21 (NOT verified)	R6′ 4.4 m on B20 / a new full-length B21′: the same transfer, ≈ 27 % less load; the office roof spans E–W; a light roof possible. X5 open
Clerestory light	winter noon 81 %, east window sun from ≈ 10:00	the same at noon; the east window's winter sun from ≈ 11:00
Office access	from the gallery end (family only); 20.2 m² with west and south windows	from the gym (family) and the external stair (visitors, Part 2); 15.7 m², south window (east window only without the east stair)

7 · The external stair at the south-east: east or south, open or enclosed · the site, the setbacks, the route from the gate



site plan (base.typ frame: origin M1, y along the M line) · 1 m = 0.33 cm · true north 21.4° off the page's up

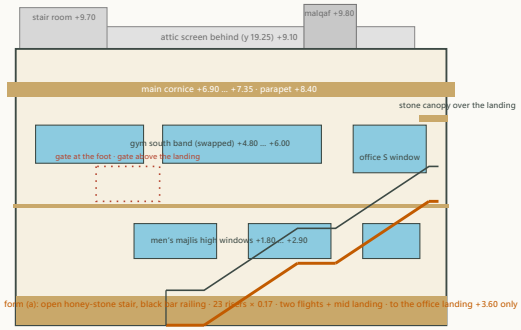
Space, setbacks and envelope at the south-east corner (real distances, site frame)

Side	to the envelope (setback line)	to the boundary	room for the stair
EAST wall face at the SE corner (y 19–24)	6.9 m	8.9 m to the lane	wide: the lane setback is tight only at the north cut (0.5 m spare); here the envelope is 6.9 m away. The enclosed tower (2.7 deep) leaves 4.2 m
SOUTH wall face (x 5–13.3)	8.2 m (no south setback: the envelope is the boundary)	8.2 m to the south boundary	the whole south garden; the nearest palm trunk 6.1 m from the SE corner

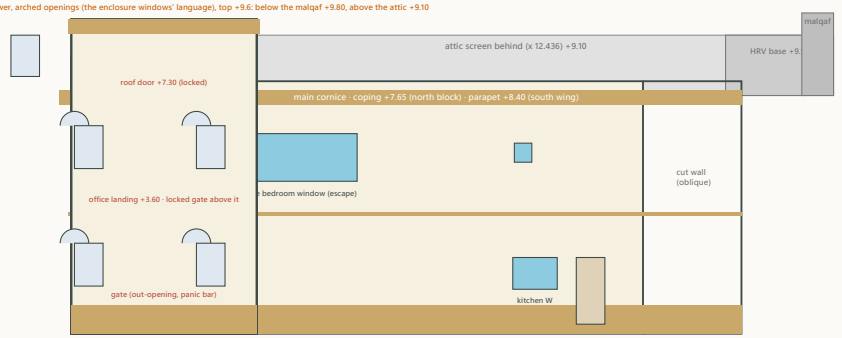
Stair footprint	spare to the envelope	to the nearest palm trunk face	ok
enclosed tower, EAST	4.1 m	3.3 m	✓
enclosed tower, SOUTH	5.5 m	6.1 m	✓
open stair, EAST	4.3 m	4.7 m	✓
open stair, SOUTH	5.5 m	6.2 m	✓

- **Route from the site gate (M1, north-west):** visitors walk down the drive on the west (where the cars park), round the south-west corner past the guest foyer, and along the south garden: ≈ 36.8 m to a south stair, ≈ 52 m round the corner to an east stair. They never enter the house. **Workers** come best through a **service gate on the lane** (assumed near the east yard, not yet in the project files): ≈ 11.3 m along the east side to an east stair, past the service yard (rubbish, deliveries, gas) and well clear of the courtyard windows.
- **Relations:** the **east** stair sits on the service side, next to the yard and the lane: workers' side, the south garden and the palms untouched; the **south** stair sits on the garden side, closer to the drive / parking (≈ 5 m shorter walk) but it crosses the men's majlis south facade in front of its high windows (sill +1.8 at the ground floor: a stair user looks straight in) and takes the garden strip where the **outdoor kitchen** would go (its position is not fixed in the project files).
- **The office's door:** on the east wall (y 21.0–21.9) for an east stair, on the south wall (x 12.0–12.9) for a south stair.

8 · The two forms in elevation · 9 · Roof access · 10 · Security, fire, maintenance route, weather, cost · 11 · Recommendation and the decisions needed



SOUTH elevation (seen from the garden), form (a) on the south · 1 m = 0.42 cm · ground −0.30



EAST elevation (seen from the lane; north on the right), form (b) on the east · 1 m = 0.42 cm

8 · Which looks better

- **(a) the open stone stair** reads as a garden stair of the Hejaz houses: honest, light, cheap; but it draws a long diagonal across a facade whose language is horizontal (the band, the cornice, the parapet) and it can only reach the office landing (a second run to the roof stacked over it has no headroom). On the south it crosses the men's majlis windows; on the east it runs under the suite bedroom's escape window. **It looks added on unless it is short and tucked into the corner.**
- **(b) the enclosed tower** with two arched openings per storey in the enclosure windows' language, a stone plinth and the main cornice line carried round it, top +9.6: a designed mass. On the **east** it pairs with the malqaf tower (+9.80) at the other end of the lane facade: two vertical elements on the service side, the attic screen between them, the garden facade untouched. On the **south** it would stand alone in the middle of the quiet garden front. **The east tower looks designed; it works best if it also replaces the internal roof stair (§ 9), so the stair room at the north-west disappears and the roofline keeps exactly two verticals.**

9 · Roof access

- **(i) the tower continues to the roof:** a locked steel gate across the stair above the office landing; the roof door at +7.30 opens onto the south wing roof (parapet +8.40). The internal roof flights and the stair room become unnecessary: **proposal** for `studies/roof.typ` (not applied): no stair room Z1 (−15.7 m² footprint, the attic's west leg straightened), no flights 3–4 (the first-floor arrival stays), the roof slab opening closed (R16–R18 become ordinary slab, the lift headroom unchanged), the walkway network starts at the south-east, the owner's "roof for maintenance only" rule is kept without any door from the roof into the house.
- **(ii) the stair serves the office only; the roof by a fixed ladder:** a caged steel ladder from the office landing (+3.60) to the roof (3.7 m) inside the tower's upper part, with a locked hatch at the top; shorter tower (+6.9, under the cornice), cheaper, less comfortable for workers with equipment (filters, PV panels). The internal roof stair then stays.

10 · Security, fire, the maintenance route, weather, cost

- **Security:** a steel gate at the foot, locked against entry (key / code outside) and opening **outwards without a key** (panic bar); a second locked gate across the stair above the office landing (visitors stop there); the office's door to the gym with a thumb-turn on the office side only; **no door from the roof into the house** (with (i) the stair room goes; the attic strip doors lead only to roof areas); cameras at the foot gate, on the office landing (the door) and at the roof arrival, plus one on the roof looking along the south wing. **The courtyard from the roof:** the sky hole's 1.10 guard stops falls, not a person with a rope: the W1 / W2 casements get key locks and the alarm's roof contacts (the roof door, the gate); nothing physical short of a grille over the sky hole, which would spoil the courtyard — not proposed.
- **Fire (second way out of the first floor):** the family reaches the stair only through the gym and the office; the office door is locked from the office side and the tower's openings sit within 1.8 m of the office window and (east) under the suite bedroom's window: **not counted** as an escape. It can be counted if the office door gets a thumb-turn on both sides and the openings near the stair are fire-rated or 1.8 m away.
- **Roof maintenance route:** from the roof door (SE) west along the south wing on paving pads (the walkway y 22.6, south of the suite court's guards: no opening crossed), through the attic door at (2.1–2.9, y 19.25) into the north block, then the existing walkway network to the water tank, the PV, the condensers (Z2b) and the HRV base with the tower (NE); no walking on the insulation; the court hole and the sky hole stay guarded.
- **Weather and dust:** the tower's arched openings unglazed with a bird mesh (rain and dust enter a little: a drained stone floor, the landing door sheltered); an open stair gets a stone canopy over the landing only. Finishes: honey-stone treads and plinth, cream plaster, black steel gates and rails.
- **Rough relative cost (orders, not quotes):** open stair (a) to the office ≈ 0.8–1.0 × the internal roof flights + stair room it does not replace; the enclosed tower (b) to the roof ≈ 1.5–2 × that, but with (i) it **replaces** the two internal flights, the stair room and the roof opening (≈ 0.6–0.8 × back), so the net extra is ≈ 0.7–1.2 ×. The QS prices it.

11 · Recommendation

- **South wing: adopt the swap.** The gym gets its real treadmill zone on the stiffest part of the slab, the office moves to the quiet wall beside the suite, the court keeps its light and gains privacy; the structural transfer (X5) gets lighter. The price: a smaller office (14 m² clear) and a smaller court (8.8 m²); the east clerestory window loses about an hour of early winter sun. Keep the court 2.0 m deep (the clerestory light); a 2.4 m court is possible at the cost of the pilates strip.
- **External stair: east, enclosed tower (b), continuing to the roof (i)** with the locked gate above the office landing, and the internal roof stair and stair room dropped as a consequence (a cleaner roofline: the malqaf and the stair tower only). Visitors come down the drive and along the south garden (≈ 52 m); workers through a service gate on the lane (to be added to the site plan).

Decisions needed from the owner

- The swap: yes / no; the court's depth (2.0 / 2.4); the pivoting TV panel (yes / no; one screen or two).
- The stair: east / south; open (a) / enclosed (b); roof by the stair (i) or by a ladder (ii); if (i): drop the internal roof flights and the stair room (then `studies/roof.typ`, `studies/structure.typ` and the 3D change).
- The service gate on the lane (position); the outdoor kitchen's position; cameras and locks as listed.

What the engineers confirm

- Structural: everything in § 4; the stair tower's foundation beside the east footings and its tie to the house; the roof slab opening (closed or kept).
- Acoustic: the pivot panel's real Rw and the gaskets; the bedroom 3 double wall.
- Fire / security consultant: the escape count, the gate hardware, the cameras.